

Linear Algebra for Many-Body Physics and Quantum Computing

Introductory Chapters

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Preface

These chapters collect and reorganise the linear algebra material from the introductory weeks of the many-body physics course. The goal is to give a self-contained, coherent treatment of the mathematical structures that underpin the rest of the course — vectors, matrices, Dirac bra-ket notation, orthonormal bases, unitary transformations, tensor products, determinants and Slater determinants — while making explicit at each step how those structures appear in many-body methods (Hartree-Fock, FCI, coupled cluster, ...) and in quantum computing (qubit states, quantum gates, quantum circuits).

The two source files (`week34.tex` and `week1.tex`) contain essentially the same linear algebra definitions, developed in parallel for the many-body course and for the quantum computing course. Here those parallel threads are woven into a single, logically ordered narrative structured as two chapters:

- **Chapter 1:** The core linear algebra — notation, vectors, matrices, Dirac notation, special matrices, algebraic operations, unitary transformations, tensor products, spectral decomposition.
- **Chapter 2:** Applications to many-body physics and quantum computing — the many-body Hamiltonian, the Pauli principle, Slater determinants, the anti-symmetriser, basis changes, the Hartree-Fock energy functional, the exponential-dimension problem, quantum gates, and an overview of how linear algebra links all many-body methods.

Shaded boxes like the one below signal a direct connection to either many-body physics or quantum computing:

Connection. This is where the abstract linear algebra concept just introduced reappears concretely in a many-body method or in a quantum circuit.

1 Linear Algebra: Notation, Vectors, Matrices and Operations

1.1 Notations and conventions

Throughout these notes vectors, matrices and higher-order tensors are always set in boldface. Vectors are denoted by lower-case letters ($\mathbf{x}, \mathbf{v}, \dots$) and matrices and tensors by upper-case letters ($\mathbf{A}, \mathbf{U}, \dots$).

Unless stated otherwise the elements v_i of a vector $\mathbf{v}$ are assumed real. A real vector of length n satisfies $\mathbf{x} \in \mathbb{R}^n$; for complex vectors we write $\mathbf{x} \in \mathbb{C}^n$. A real $n \times n$ matrix satisfies $\mathbf{A} \in \mathbb{R}^{n \times n}$. Index counting starts at zero: matrix elements run from a_{00} to $a_{n-1, n-1}$.

We use the standard shorthand symbols: $\forall$ (for all), $\Rightarrow$ (implies), $\equiv$ (equivalent / defined as), with $\mathbb{R}, \mathbb{I}, \mathbb{C}$ for the real, integer and complex number fields.

1.2 Vectors

A column vector of length n and its transpose are

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}, \quad \mathbf{x}^T = [x_0 \ x_1 \ \cdots \ x_{n-1}].$$

For complex vectors the *Hermitian conjugate* (adjoint) is

$$\mathbf{x}^\dagger = [x_0^* \ x_1^* \ \cdots \ x_{n-1}^*].$$

The *inner (dot) product* is the scalar

$$\mathbf{x}^T \mathbf{x} = \sum_{i=0}^{n-1} x_i x_i = x_0^2 + x_1^2 + \cdots + x_{n-1}^2,$$

and for complex vectors $\mathbf{x}^\dagger \mathbf{x} = \sum_{i=0}^{n-1} x_i^* x_i = \|\mathbf{x}\|^2$.

The *outer product* of two real vectors $\mathbf{y}$ (length n) and $\mathbf{z}$ (length m) is the $n \times m$ matrix

$$\mathbf{y} \mathbf{z}^T \Rightarrow (\mathbf{y} \mathbf{z}^T)_{ij} = y_i z_j.$$

For complex vectors one writes $\mathbf{y} \mathbf{z}^\dagger$ with $(\mathbf{y} \mathbf{z}^\dagger)_{ij} = y_i z_j^*$.

Important vector operations:

Addition / subtraction: $\mathbf{x} = \mathbf{y} \pm \mathbf{z} \Rightarrow x_i = y_i \pm z_i$.

Scalar multiplication: $\mathbf{x} = \gamma \mathbf{y} \Rightarrow x_i = \gamma y_i$.

Hadamard (element-wise) product: $\mathbf{x} = \mathbf{y} \circ \mathbf{z} \Rightarrow x_i = y_i z_i$.

1.3 Matrices: definitions and algebraic operations

A general $n \times n$ matrix and its column-vector representation:

$$\mathbf{A} = \begin{bmatrix} a_{00} & a_{01} & \cdots & a_{0,n-1} \\ a_{10} & a_{11} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & \cdots & \cdots & a_{n-1,n-1} \end{bmatrix} = [\mathbf{a}_0 \quad \mathbf{a}_1 \quad \cdots \quad \mathbf{a}_{n-1}].$$

The fundamental algebraic operations are:

$$\text{Scalar mult.: } \mathbf{A} = \gamma \mathbf{B} \Rightarrow a_{ij} = \gamma b_{ij}.$$

$$\text{Addition: } \mathbf{A} = \mathbf{B} \pm \mathbf{C} \Rightarrow a_{ij} = b_{ij} \pm c_{ij}.$$

$$\text{Matrix-vector product: } \mathbf{y} = \mathbf{A}\mathbf{x} \Rightarrow y_i = \sum_{j=0}^{n-1} a_{ij} x_j.$$

$$\text{Matrix-matrix product: } \mathbf{A} = \mathbf{B}\mathbf{C} \Rightarrow a_{ij} = \sum_{k=0}^{n-1} b_{ik} c_{kj}.$$

$$\text{Transpose: } \mathbf{A} = \mathbf{B}^T \Rightarrow a_{ij} = b_{ji}.$$

Inverse. If $\mathbf{A}$ is square and nonsingular, its inverse satisfies $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix.

Hermitian conjugate. The Hermitian conjugate $\mathbf{A}^\dagger$ is obtained by transposing $\mathbf{A}$ and complex-conjugating every element: $(A^\dagger)_{ij} = a_{ji}^*$.

Nonsingularity: equivalent characterisations. For an $n \times n$ matrix $\mathbf{A}$ the following properties are all equivalent: (i) $\mathbf{A}^{-1}$ exists; (ii) $\mathbf{A}\mathbf{x} = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$; (iii) the rows of $\mathbf{A}$ form a basis of $\mathbb{C}^n$; (iv) the columns of $\mathbf{A}$ form a basis of $\mathbb{C}^n$; (v) $\mathbf{A}$ is a product of elementary matrices; (vi) zero is not an eigenvalue of $\mathbf{A}$.

1.4 Special matrix types

Table 1 lists the matrix types that appear most frequently in quantum mechanics and many-body physics.

Structured (sparse) matrices important for numerical many-body work:

- **Diagonal:** $a_{ij} = 0$ for $i \neq j$.
- **Upper/lower triangular:** $a_{ij} = 0$ for $i > j$ / $i < j$.
- **Upper/lower Hessenberg:** $a_{ij} = 0$ for $i > j + 1$ / $i < j + 1$.
- **Tridiagonal:** $a_{ij} = 0$ for $|i - j| > 1$.
- **Banded with bandwidth p :** $a_{ij} = 0$ for $|i - j| > p$.
- Block upper/lower triangular matrices, and more.

Name	Defining relation	Element condition
Symmetric	$\mathbf{A} = \mathbf{A}^T$	$a_{ij} = a_{ji}$
Real orthogonal	$\mathbf{A} = (\mathbf{A}^T)^{-1}$	$\sum_k a_{ik} a_{jk} = \delta_{ij}$
Hermitian	$\mathbf{A} = \mathbf{A}^\dagger$	$a_{ij} = a_{ji}^*$
Unitary	$\mathbf{A} = (\mathbf{A}^\dagger)^{-1}$	$\sum_k a_{ik} a_{jk}^* = \delta_{ij}$
Real matrix	$\mathbf{A} = \mathbf{A}^*$	$a_{ij} = a_{ij}^*$

Table 1: Important special matrix types and their defining properties.

Many-body connection. Hamiltonian matrices in shell-model (full configuration-interaction, FCI) calculations are typically large, sparse, and often banded. Exploiting sparsity is essential to diagonalise them for systems with many single-particle states.

1.5 Dirac bra-ket notation

In quantum mechanics it is conventional to write state vectors using Dirac's bra-ket formalism. A ket $|x\rangle$ is a column vector

$$|x\rangle = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix},$$

and the corresponding bra $\langle x|$ is the row vector (dual)

$$\langle x| = [x_0^* \quad x_1^* \quad \cdots \quad x_{n-1}^*],$$

so $|x\rangle^\dagger = \langle x|$. The *inner product* (a complex number) is

$$\langle y|x\rangle = \sum_{i=0}^{n-1} y_i^* x_i.$$

Note that $\langle x|x\rangle \geq 0$ is always real and non-negative, and $\langle y|x\rangle^* = \langle x|y\rangle$.

The *outer product* $|x\rangle\langle y|$ is an $n \times n$ matrix with elements $(|x\rangle\langle y|)_{ij} = x_i y_j^*$:

$$|x\rangle\langle y| = \begin{bmatrix} x_0 y_0^* & x_0 y_1^* & \cdots & x_0 y_{n-1}^* \\ x_1 y_0^* & x_1 y_1^* & \cdots & x_1 y_{n-1}^* \\ \vdots & \vdots & \ddots & \vdots \\ x_{n-1} y_0^* & \cdots & \cdots & x_{n-1} y_{n-1}^* \end{bmatrix}.$$

Worked example. Let $|x\rangle = \begin{bmatrix} 1 - i \\ 2 + i \end{bmatrix}$. Then

$$\langle x|x\rangle = (1 + i)(1 - i) + (2 - i)(2 + i) = 2 + 5 = 7 \in \mathbb{R}.$$

For two different vectors $|x\rangle = \begin{bmatrix} -1 \\ 2i \\ 1 \end{bmatrix}$, $|y\rangle = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$: $\langle x|y\rangle = -1 + i \neq \langle y|x\rangle = -1 - i$, confirming the rule $\langle x|y\rangle^* = \langle y|x\rangle$.

1.6 Computational basis states and superposition

A *two-level system* (qubit) is described by two orthonormal basis states, which we label

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \langle 0|1\rangle = 0, \quad \langle 0|0\rangle = \langle 1|1\rangle = 1.$$

These are the *computational basis* states. A general state is a *superposition*

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha = \langle 0|\psi\rangle$ and $\beta = \langle 1|\psi\rangle$ are the probability amplitudes, satisfying $|\alpha|^2 + |\beta|^2 = 1$.

The identity operator decomposes as a sum of outer products:

$$\mathbf{I} = \sum_{i=0}^1 |i\rangle\langle i| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Projection operators. The outer products

$$\mathbf{P} = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{Q} = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

are *projection operators*. They are *idempotent* ($\mathbf{P}^2 = \mathbf{P}$, $\mathbf{Q}^2 = \mathbf{Q}$), commute with each other ($[\mathbf{P}, \mathbf{Q}] = 0$), and satisfy $\mathbf{PQ} = 0$, $\mathbf{P} + \mathbf{Q} = \mathbf{I}$. Acting on a superposition state:

$$\mathbf{P}|\psi\rangle = \alpha|0\rangle, \quad \mathbf{Q}|\psi\rangle = \beta|1\rangle.$$

The measurement postulate states that a measurement on $|\psi\rangle$ yields $|0\rangle$ with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$.

Many-body connection. In many-body theory, projection operators $\hat{P}$ (onto occupied states) and $\hat{Q}$ (onto unoccupied / virtual states) define the particle-hole decomposition that is fundamental to perturbation theory, Hartree-Fock theory, and coupled-cluster methods.

Density matrix. The density matrix (density operator) of a pure state $|\psi\rangle$ is the outer product

$$\boldsymbol{\rho} = |\psi\rangle\langle\psi| = \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix},$$

with $\text{tr } \boldsymbol{\rho} = |\alpha|^2 + |\beta|^2 = 1$. Density matrices generalise to mixed states and subsystems; they are central to quantum information theory and to reduced-density-matrix many-body methods.

1.7 Pauli matrices

The three Pauli matrices are

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Key properties (easy to verify directly):

- **Involutory:** $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \mathbf{I}$.
- **Unitary:** $\sigma_\alpha^\dagger = \sigma_\alpha^{-1}$ (Hermitian conjugate equals inverse).
- **Hermitian:** $\sigma_\alpha^\dagger = \sigma_\alpha$.
- $\det(\sigma_\alpha) = -1$ for each α .
- **Commutation:** $[\sigma_x, \sigma_y] = 2i\sigma_z$ (and cyclic permutations).

The Pauli matrices act on the computational basis as

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle; \quad \sigma_z|0\rangle = +|0\rangle, \quad \sigma_z|1\rangle = -|1\rangle.$$

Acting on a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

$$\sigma_x|\psi\rangle = \beta|0\rangle + \alpha|1\rangle, \quad \sigma_y|\psi\rangle = i\beta|0\rangle - i\alpha|1\rangle.$$

Quantum computing connection. The Pauli matrices are the elementary single-qubit gates X , Y , Z . Together with the Hadamard gate $H = (\sigma_x + \sigma_z)/\sqrt{2}$ and the phase gate T they form a universal single-qubit gate set.

Many-body connection. Via the Jordan-Wigner transformation, the creation/annihilation operators of second quantisation are mapped directly onto products (strings) of Pauli matrices, enabling the representation of fermionic Hamiltonians on qubit registers.

1.8 Orthonormal bases and the completeness relation

A set of states $\{|\phi_i\rangle\}_{i=0}^{n-1}$ is an *orthonormal basis* (ONB) of an n -dimensional Hilbert space if

$$\langle\phi_j|\phi_i\rangle = \delta_{ij} \quad \text{and} \quad \sum_{i=0}^{n-1} |\phi_i\rangle\langle\phi_i| = \mathbf{I} \quad (\text{completeness / resolution of the identity}).$$

Any state can then be expanded as $|\psi\rangle = \sum_i \langle\phi_i|\psi\rangle |\phi_i\rangle$, with $\langle\phi_i|\psi\rangle$ the projection (overlap) of $|\psi\rangle$ onto $|\phi_i\rangle$.

Many-body connection. The natural ONB in many-body physics is the set of single-particle eigenstates of the one-body Hamiltonian:

$$\hat{h}_0|\phi_i\rangle = \varepsilon_i|\phi_i\rangle, \quad \phi_i(\mathbf{r}) = \langle\mathbf{r}|\phi_i\rangle.$$

This single-particle basis $\{|\phi_i\rangle\}$ is the starting point for virtually all many-body

methods: Hartree-Fock, MBPT, FCI, coupled-cluster, and quantum Monte Carlo.

1.9 Unitary transformations and preservation of orthogonality

An $n \times n$ matrix $\mathbf{U}$ is *unitary* if $\mathbf{U}^\dagger \mathbf{U} = \mathbf{U} \mathbf{U}^\dagger = \mathbf{I}$, which means $\mathbf{U}^{-1} = \mathbf{U}^\dagger$.

Orthogonality is preserved. If $\{\mathbf{v}_i\}$ is an orthonormal set, $\mathbf{v}_j^T \mathbf{v}_i = \delta_{ij}$, then the transformed set $\mathbf{w}_i = \mathbf{U} \mathbf{v}_i$ is also orthonormal:

$$\mathbf{w}_j^T \mathbf{w}_i = (\mathbf{U} \mathbf{v}_j)^T \mathbf{U} \mathbf{v}_i = \mathbf{v}_j^T \underbrace{\mathbf{U}^T \mathbf{U}}_{=\mathbf{I}} \mathbf{v}_i = \mathbf{v}_j^T \mathbf{v}_i = \delta_{ij}.$$

In Dirac notation, if $|\psi_i\rangle = \sum_j u_{ij} |\phi_j\rangle$ with (u_{ij}) unitary, then $\langle \psi_j | \psi_i \rangle = \delta_{ij}$.

Why only unitary transformations? For the inner product (and hence the norm) to be preserved under $|\phi\rangle = \mathbf{U} |\psi\rangle$ we need

$$\langle \phi | \phi \rangle = \langle \psi | \mathbf{U}^\dagger \mathbf{U} | \psi \rangle = \langle \psi | \psi \rangle = 1,$$

which requires $\mathbf{U}^\dagger \mathbf{U} = \mathbf{I}$. Since probabilities must sum to unity, this is a fundamental physical requirement. Unitary transformations are rotations (and reflections) in Hilbert space. The fact that $\mathbf{U}^{-1} = \mathbf{U}^\dagger$ also means that every unitary transformation is perfectly reversible.

Example: two-state rotation. Given $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and a unitary $\mathbf{U}$, the rotated state $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle = \mathbf{U} |\psi\rangle$ satisfies

$$\begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Since $|\alpha|^2 + |\beta|^2 = 1$ and $\mathbf{U}$ is unitary, $|\gamma|^2 + |\delta|^2 = 1$ automatically.

Many-body connection. In Hartree-Fock theory the self-consistent field (SCF) procedure generates a sequence of unitary rotations of the single-particle basis. The total HF energy is invariant under unitary rotations within the occupied or virtual subspace separately, which is the Thouless theorem.

Quantum computing connection. Every quantum gate is a unitary matrix. The evolution of a closed quantum system is described entirely by unitary operations; this is the quantum analogue of classical reversibility.

1.10 Tensor products

The *tensor (Kronecker) product* of two vectors $|x\rangle$ (length n) and $|y\rangle$ (length m) is the vector of length nm :

$$|x\rangle \otimes |y\rangle = |xy\rangle = \begin{bmatrix} x_0 y_0 \\ x_0 y_1 \\ x_1 y_0 \\ x_1 y_1 \end{bmatrix} \quad (\text{for } n = m = 2).$$

For the computational basis:

$$|0\rangle\otimes|0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad |0\rangle\otimes|1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad |1\rangle\otimes|0\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad |1\rangle\otimes|1\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

These four states form an ONB for the two-qubit Hilbert space $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$.

The most general state of N qubits is

$$|\Psi\rangle = \sum_{i_1 i_2 \dots i_N} \Psi_{i_1 i_2 \dots i_N} |i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle,$$

requiring 2^N complex amplitudes. For $N \approx 300$ this exceeds the number of atoms in the observable universe — this is the *exponential complexity* at the heart of quantum many-body physics.

Many-body connection. The full N -particle Hilbert space is a tensor product of N single-particle spaces. The FCI method works in this exponentially large space; all approximate methods (HF, MBPT, coupled cluster, DMRG, ...) are strategies to identify the relevant low-dimensional submanifold.

Quantum computing connection. A quantum computer with N qubits *inherently* represents a 2^N -dimensional state vector. This is both the source of its potential power and the reason that quantum simulation of many-body systems is a leading application.

1.11 Spectral decomposition of Hermitian operators

A Hermitian operator $\mathbf{A} = \mathbf{A}^\dagger$ has the following key properties:

1. All eigenvalues λ_i are *real*.
2. Eigenvectors belonging to distinct eigenvalues are *orthogonal*.
3. $\mathbf{A}$ is diagonalisable (it is a *normal* operator, $[\mathbf{A}, \mathbf{A}^\dagger] = 0$).

The eigenvalue equation $\mathbf{A}|\psi\rangle = \lambda|\psi\rangle$ has non-trivial solutions only when

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \quad (\text{characteristic polynomial}).$$

Spectral decomposition. Let $\{|i\rangle\}$ be the eigenbasis of $\mathbf{A}$ with eigenvalues λ_i . Any state expands as $|\psi\rangle = \sum_i \alpha_i |i\rangle$. Using projection operators $\mathbf{P}_i = |i\rangle\langle i|$ we obtain:

$$\mathbf{A}|\psi\rangle = \sum_i \alpha_i \lambda_i |i\rangle = \sum_i \lambda_i \mathbf{P}_i |\psi\rangle,$$

from which, since this holds for any $|\psi\rangle$,

$$\boxed{\mathbf{A} = \sum_{i=0}^{n-1} \lambda_i |i\rangle\langle i| = \sum_{i=0}^{n-1} \lambda_i \mathbf{P}_i.}$$

This is the *spectral decomposition*: the operator is expressed as a weighted sum of its projection operators.

Example for a 2×2 operator. If $\mathbf{A}$ has eigenstates $|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $|\psi_b\rangle = \beta_0|0\rangle + \beta_1|1\rangle$ with eigenvalues λ_a, λ_b , then

$$\mathbf{A} = \lambda_a \begin{bmatrix} |\alpha_0|^2 & \alpha_0\alpha_1^* \\ \alpha_1\alpha_0^* & |\alpha_1|^2 \end{bmatrix} + \lambda_b \begin{bmatrix} |\beta_0|^2 & \beta_0\beta_1^* \\ \beta_1\beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

Many-body connection. The spectral decomposition is used every time one diagonalises a Hamiltonian matrix. In FCI the full N -body Hamiltonian is diagonalised in the many-determinant basis; in Hartree-Fock the Fock matrix (a single-particle Hermitian operator) is diagonalised to find the optimal orbital basis.

Quantum computing connection. Any unitary gate $\mathbf{U}$ can be written as $\mathbf{U} = e^{i\mathbf{H}}$ for a Hermitian generator $\mathbf{H}$. The spectral decomposition of $\mathbf{H}$ gives $\mathbf{U} = \sum_i e^{i\lambda_i} |i\rangle\langle i|$. This exponential relationship connects the physical Hamiltonian to the time-evolution operator $\hat{U} = e^{-i\hat{H}t/\hbar}$ and is the basis of the variational quantum eigensolver (VQE) and unitary coupled-cluster (UCC) ansatz.

2 From Linear Algebra to Many-Body Physics and Quantum Computing

2.1 The many-body Hamiltonian

Throughout these lectures we assume that the interacting part of the Hamiltonian can be approximated by a two-body interaction, so the full N -particle Hamiltonian takes the form

$$\hat{H} = \hat{H}_0 + \hat{H}_I = \sum_{i=1}^N \hat{h}_0(x_i) + \sum_{i<j}^N \hat{v}(r_{ij}), \quad (1)$$

where x_i denotes the coordinates (position and spin) of particle i , and $\hat{h}_0$ is the one-body part (kinetic energy plus external potential). The one-body potential is typically a harmonic oscillator, a Coulomb interaction, or the self-consistent Hartree-Fock potential.

The Hamiltonian (1) is *symmetric* under permutation of any two particles: for any permutation operator $\hat{P}_{ij}$, $[\hat{H}, \hat{P}_{ij}] = 0$.

2.2 The Pauli exclusion principle and antisymmetry

Since we deal with fermions, the total wave function must be antisymmetric under interchange of any two particles. The permutation operator $\hat{P}_{ij}$ interchanges particles i and j :

$$\hat{P}_{ij}\Psi(x_1, \dots, x_i, \dots, x_j, \dots, x_N) = \beta \Psi(x_1, \dots, x_i, \dots, x_j, \dots, x_N).$$

The Pauli principle requires $\beta = -1$: the fermionic wave function changes sign under every transposition.

The *antisymmetrisation operator* is

$$\hat{A} = \frac{1}{N!} \sum_p (-1)^p \hat{P},$$

where p counts the parity of each permutation. It satisfies

$$\hat{A}^2 = \hat{A} \quad (\text{idempotency}), \quad [\hat{H}_0, \hat{A}] = [\hat{H}_I, \hat{A}] = 0. \quad (2)$$

2.3 Determinants

For an $n \times n$ matrix $\mathbf{A}$, the determinant is defined by summing over all $n!$ permutations of column indices:

$$\det(\mathbf{A}) = |\mathbf{A}| = \sum_{i=1}^{n!} (-1)^{p_i} \hat{P}_i a_{11} a_{22} \cdots a_{nn},$$

where p_i is the number of transpositions needed to restore the natural ordering. For $n = 2$:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

Linearity in a column. A fundamental property used repeatedly in Hartree-Fock theory is the linearity of the determinant in any single column:

$$\begin{vmatrix} a_{11} & \cdots & \sum_{k=1}^n c_k b_{1k} & \cdots \\ a_{21} & \cdots & \sum_{k=1}^n c_k b_{2k} & \cdots \\ \vdots & & \vdots & \\ a_{n1} & \cdots & \sum_{k=1}^n c_k b_{nk} & \cdots \end{vmatrix} = \sum_{k=1}^n c_k \begin{vmatrix} a_{11} & \cdots & b_{1k} & \cdots \\ a_{21} & \cdots & b_{2k} & \cdots \\ \vdots & & \vdots & \\ a_{n1} & \cdots & b_{nk} & \cdots \end{vmatrix}.$$

Product rule. If every column of $\mathbf{A}$ is a linear combination of columns of $\mathbf{B}$, with coefficient matrix $\mathbf{C}$ (i.e. $a_{ij} = \sum_k b_{ik} c_{kj}$, meaning $\mathbf{A} = \mathbf{B}\mathbf{C}$), then

$$\det(\mathbf{A}) = \det(\mathbf{B}) \cdot \det(\mathbf{C}).$$

Verify this for $n = 2$ as an exercise.

2.4 The Slater determinant

The antisymmetric N -particle wave function built from single-particle orbitals $\psi_\alpha, \psi_\beta, \dots, \psi_\sigma$ is the *Slater determinant*

$$\Phi(x_1, \dots, x_N; \alpha, \dots, \sigma) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_\alpha(x_1) & \psi_\alpha(x_2) & \cdots & \psi_\alpha(x_N) \\ \psi_\beta(x_1) & \psi_\beta(x_2) & \cdots & \psi_\beta(x_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_\sigma(x_1) & \psi_\sigma(x_2) & \cdots & \psi_\sigma(x_N) \end{vmatrix}, \quad (3)$$

where x_i encodes both spatial coordinates and spin of particle i .

The Slater determinant automatically enforces:

- Antisymmetry under particle exchange (Pauli principle): swapping two rows reverses the sign.
- No two fermions in the same state: two identical rows give $\det = 0$.

An equivalent expression using the antisymmetriser is $\Phi = \sqrt{N!} \hat{A} \Phi_H$, where $\Phi_H = \psi_\alpha(x_1) \psi_\beta(x_2) \cdots \psi_\sigma(x_N)$ is the (unsymmetrised) *Hartree product*.

2.5 Basis changes and the Slater determinant under unitary rotations

It is standard practice to expand the single-particle functions in a fixed orthogonal basis $\{\phi_\lambda\}$ and vary the coefficients. The new orbitals are

$$\psi_p^{\text{new}}(x) = \sum_{\lambda} C_{p\lambda} \phi_\lambda(x). \quad (4)$$

If the matrix $(C_{p\lambda})$ is unitary, the new basis is automatically orthonormal. In bra-ket notation:

$$|p\rangle = \sum_{\lambda} C_{p\lambda} |\lambda\rangle, \quad \langle p|q\rangle = \delta_{pq}.$$

The Slater determinant in the new basis becomes

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_1) & \cdots \\ \vdots & \ddots \end{vmatrix} = \det(\mathbf{C}) \cdot \det(\mathbf{\Phi}),$$

by the determinant product rule. Since $\mathbf{C}$ is unitary, $|\det(\mathbf{C})| = 1$, so the new Slater determinant differs from the old one only by an unobservable global phase.

Many-body connection. This is the algebraic foundation of the freedom to perform unitary rotations within the occupied subspace in Hartree-Fock theory without changing the many-body state or the total energy. It also underlies the equivalence of different choices of occupied orbitals in CI and coupled-cluster calculations.

2.6 Expectation values with a Slater determinant

Using $\hat{A}^2 = \hat{A}$ and orthogonality of the single-particle functions, the expectation value of the Hamiltonian (1) with the Slater determinant (3) factorises cleanly.

One-body contribution.

$$\langle \Phi | \hat{H}_0 | \Phi \rangle = \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle, \quad \text{where} \quad \langle \mu | \hat{h}_0 | \mu \rangle = \int \psi_{\mu}^*(x) \hat{h}_0 \psi_{\mu}(x) dx.$$

Two-body contribution.

$$\begin{aligned} \langle \Phi | \hat{H}_I | \Phi \rangle = \frac{1}{2} \sum_{\mu=1}^N \sum_{\nu=1}^N & \left[\underbrace{\int \psi_{\mu}^*(x_i) \psi_{\nu}^*(x_j) \hat{v}(r_{ij}) \psi_{\mu}(x_i) \psi_{\nu}(x_j) dx_i dx_j}_{\text{direct (Hartree) term: } \langle \mu\nu | \hat{v} | \mu\nu \rangle} \right. \\ & \left. - \underbrace{\int \psi_{\mu}^*(x_i) \psi_{\nu}^*(x_j) \hat{v}(r_{ij}) \psi_{\nu}(x_i) \psi_{\mu}(x_j) dx_i dx_j}_{\text{exchange (Fock) term: } \langle \mu\nu | \hat{v} | \nu\mu \rangle} \right]. \end{aligned}$$

The antisymmetrised two-body matrix element is $\langle \mu\nu | \hat{v} | \mu\nu \rangle_{\text{AS}} = \langle \mu\nu | \hat{v} | \mu\nu \rangle - \langle \mu\nu | \hat{v} | \nu\mu \rangle$.

The Hartree-Fock energy functional. Combining both contributions:

$$E[\Phi] = \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle + \frac{1}{2} \sum_{\mu, \nu=1}^N \langle \mu\nu | \hat{v} | \mu\nu \rangle_{\text{AS}}. \quad (5)$$

Using the expansion (4), this becomes

$$E[\Psi] = \sum_{i=1}^N \sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \frac{1}{2} \sum_{i,j} \sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{\text{AS}}. \quad (6)$$

Minimising (6) with respect to the coefficients ($C_{p\lambda}$), subject to the unitarity constraint, yields the Hartree-Fock equations. From a linear algebra standpoint, these are a self-consistent eigenvalue problem: diagonalise the Fock matrix (which itself depends on the solution through the density matrix), then iterate until convergence.

Summary. The entire Hartree-Fock method is a story in linear algebra: expand states in an ONB (Section 1.8), apply a unitary transformation (Section 1.9), use the product rule for determinants (Section 2.3), and solve a Hermitian eigenvalue problem (Section 1.11).

2.7 Quantum computing: the gate model and its linear-algebra foundation

A quantum computer is a well-controlled quantum many-body system. The abstract *gate-based* model is defined as follows. Starting from N qubits initialised in $|\Psi\rangle^{(0)} = |00\cdots 0\rangle$, the state evolves by successive applications of unitary gates:

$$|\Psi\rangle^{(n+1)} = \hat{U}^{(n)}|\Psi\rangle^{(n)}, \quad \hat{U}^{(n)} = T \exp\left(-i \int_{t_n}^{t_{n+1}} \hat{H}(t) dt\right).$$

A *single-qubit gate* acts on qubit a via a 2×2 unitary:

$$\Psi_{i_1 \cdots i_N}^{(n+1)} = \sum_{i'_a} U_{i_a i'_a}^{(n)} \Psi_{i_1 \cdots i_{a-1} i'_a i_{a+1} \cdots i_N}^{(n)}.$$

A *two-qubit gate* acts on qubits a, b via a 4×4 unitary. Standard examples are the Pauli gates X, Y, Z (see Section 1.7), the Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

the phase gate $T = \begin{pmatrix} e^{i\pi/8} & 0 \\ 0 & e^{-i\pi/8} \end{pmatrix}$, and the two-qubit CNOT (controlled-NOT):

$$C_X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

When qubit a is *measured*, the result $\alpha \in \{0, 1\}$ occurs with probability

$$P_\alpha = \sum_{i_1 \cdots i_a \cdots i_N} \left| \Psi_{i_1 \cdots \alpha \cdots i_N}^{(n)} \right|^2,$$

after which the state collapses to the corresponding projected state.

In a nutshell. Quantum computing is a restricted form of linear algebra: matrix-vector multiplications with special unitary matrices (gates) on exponentially large vectors (wave functions). The challenge — shared with classical many-body physics — is that only a tiny fraction of the 2^N -dimensional space is physically relevant; identifying that subspace is the core problem in both fields.

2.8 The exponential-dimension problem and why it matters

The state of N spin- $\frac{1}{2}$ particles requires 2^N complex numbers to describe. For $N \approx 300$, $2^{300} \gg 10^{82}$ (the estimated number of atoms in the observable universe).

How do physicists actually compute?

1. **Clever approximations:** Bethe ansatz, perturbation theory, mean-field approximations.
2. **Computer simulations:** DFT, quantum Monte Carlo, FCI, coupled cluster, DMRG.
3. **Restricting to tractable subsystems:** systems where a single-particle basis is a good approximation.

For a quantum circuit of depth D acting on N qubits, the accessible subspace has dimension $O(DN)$, vastly smaller than 2^N . For $N = 100$ and $D = 10^6$, this subspace still has 10^{20} times fewer degrees of freedom than the full Hilbert space. The question is whether the *physically relevant* states lie in this reachable subspace — a central open problem in quantum many-body theory and quantum computing alike (see, e.g., LaRocca et al.).

The variational quantum eigensolver (VQE) algorithm tackles this problem by using a parametrised quantum circuit as an ansatz for the ground state, minimising the expectation value of the Hamiltonian classically. This is the quantum-computing analogue of the variational principle that underlies Hartree-Fock and coupled-cluster theory: the ground state energy satisfies

$$E_0 \leq E[\Phi] = \int \Phi^* \hat{H} \Phi \, d\tau$$

for any normalised trial state Φ , with equality only at the exact ground state.

3 Exercises

Chapter 1: Linear algebra fundamentals

1. **Inner and outer products.** Let $\mathbf{u} = [1 - i \quad 2 + i]^T$ and $\mathbf{v} = [1 \quad i]^T$. (a) Compute $\mathbf{u}^\dagger \mathbf{v}$ and $\mathbf{v}^\dagger \mathbf{u}$. (b) Verify $(\mathbf{u}^\dagger \mathbf{v})^* = \mathbf{v}^\dagger \mathbf{u}$. (c) Compute the outer product $\mathbf{u} \mathbf{v}^\dagger$ and show $\text{tr}(\mathbf{u} \mathbf{v}^\dagger) = \mathbf{v}^\dagger \mathbf{u}$.
2. **Special matrix types.** For each matrix below, identify which type(s) from Table 1 it belongs to, and verify your answer using the element conditions in the table: (a) σ_x ; (b) σ_z ; (c) the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.
3. **Pauli algebra.** (a) Prove $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \mathbf{I}$. (b) Compute all commutators $[\sigma_i, \sigma_j]$ for $i, j \in \{x, y, z\}$. (c) Which Pauli matrix has $|0\rangle$ and $|1\rangle$ as its eigenbasis? What are the eigenvalues?
4. **Projection operators.** (a) Verify that $\mathbf{P} = |0\rangle\langle 0|$ satisfies $\mathbf{P}^2 = \mathbf{P}$ and $\det(\mathbf{P}) = 0$. (b) Show $\mathbf{P} + \mathbf{Q} = \mathbf{I}$ and $\mathbf{P}\mathbf{Q} = 0$. (c) What does $\mathbf{P}|\psi\rangle$ equal for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?
5. **Commutator identities.** Prove the following for arbitrary operators $\hat{A}$, $\hat{B}$, $\hat{C}$: (a) $[\hat{A} + \hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}]$; (b) $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$; (c) $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$ (Jacobi identity).
6. **Shared eigenstates.** Prove that if $[\hat{A}, \hat{B}] = 0$ then $\hat{A}$ and $\hat{B}$ share a common eigenbasis.

7. **Tensor products.** (a) Write out all four two-qubit basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ as column vectors of length 4 and verify they are mutually orthogonal. (b) Compute $C_X|00\rangle$, $C_X|01\rangle$, $C_X|10\rangle$, $C_X|11\rangle$. Explain the action of the CNOT gate in words.
8. **Unitary transformation.** Let $\mathbf{U}$ be a 2×2 unitary matrix and $\{|v_0\rangle, |v_1\rangle\}$ an ONB. Prove that $\{\mathbf{U}|v_0\rangle, \mathbf{U}|v_1\rangle\}$ is also an ONB.
9. **Spectral decomposition.** (a) Find the eigenvalues and normalised eigenvectors of σ_x . (b) Write the spectral decomposition $\sigma_x = \sum_i \lambda_i |i\rangle\langle i|$ explicitly. (c) Verify that $\sigma_x|\psi_a\rangle = \lambda_a|\psi_a\rangle$ for each eigenvector.

Chapter 2: Many-body physics and quantum computing

1. **Determinant product rule.** For 2×2 matrices $\mathbf{B}$ and $\mathbf{C}$, define $a_{ij} = \sum_k b_{ik} c_{kj}$. Show by direct calculation that $\det(\mathbf{A}) = \det(\mathbf{B}) \det(\mathbf{C})$.
2. **Slater determinant:** $N = 2$. Write the Slater determinant for two electrons in orbitals ψ_α and ψ_β . (a) Show it is antisymmetric under particle exchange. (b) Show it vanishes if $\psi_\alpha = \psi_\beta$.
3. **Antisymmetriser.** Verify for $N = 2$ that $\hat{A}^2 = \hat{A}$ and that $[\hat{H}_0, \hat{A}] = 0$ when $\hat{H}_0 = \hat{h}_0(x_1) + \hat{h}_0(x_2)$.
4. **Basis change of the Slater determinant.** With the expansion $\psi_p = \sum_\lambda C_{p\lambda} \phi_\lambda$ and a 2×2 unitary $\mathbf{C}$, show that the two-particle Slater determinant in the new basis equals $\det(\mathbf{C}) \det(\Phi)$ and that $|\det(\mathbf{C})| = 1$ leaves the physics unchanged.
5. **Hartree-Fock energy.** For a two-electron system with orbitals ψ_α and ψ_β , write out the HF energy functional (5) explicitly, identifying the kinetic, direct (Hartree), and exchange (Fock) contributions.
6. **Variational principle.** Show that $E[\Phi] \geq E_0$ for any normalised trial state Φ , where E_0 is the exact ground state energy. Use the spectral decomposition of $\hat{H}$ in its eigenbasis.
7. **Exponential dimension.** How many complex numbers are needed to store the full N -particle wave function for $N = 10, 20, 50$? A modern GPU can perform roughly 10^{14} floating-point operations per second. Estimate how long a single matrix-vector multiplication in the $N = 50$ space would take.
8. **One-qubit gates numerically.** Write a short Python function that represents $|0\rangle$ and $|1\rangle$ as NumPy arrays and applies each Pauli matrix, the Hadamard gate H , and the phase gate T in turn. Verify the action of X (spin flip), Z (phase flip) and $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ (creation of a superposition state).
9. **Pauli gate as a quantum NOT.** Explain why σ_x is the quantum analogue of the classical NOT gate. What is the difference between the classical and quantum cases when the input is a superposition $\alpha|0\rangle + \beta|1\rangle$?
10. **VQE and the variational principle.** The variational quantum eigensolver minimises $E(\boldsymbol{\theta}) = \langle \Psi(\boldsymbol{\theta}) | \hat{H} | \Psi(\boldsymbol{\theta}) \rangle$ over circuit parameters $\boldsymbol{\theta}$. (a) Explain why $E(\boldsymbol{\theta}) \geq E_0$ for any $\boldsymbol{\theta}$. (b) Describe the connection to the Hartree-Fock variational principle. (c) What determines the quality of the approximation in VQE compared

to FCI?